

CoTEDS-Smart

Using wearable tech (e.g., smart watches and fitness trackers) to research physical and mental health during pregnancy & parenthood.

Any TEDS twins and/or coparents enrolled in the Children of TEDS study with their child(ren) are eligible to take part. We're looking for people who use a smartphone, Fitbit, Apple Watch, Garmin or Oura Ring.

You do not need to wear your device more than you normally do.

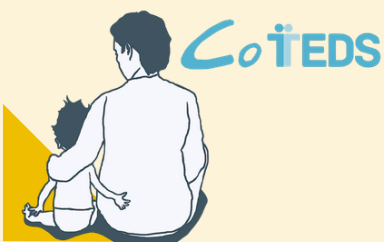
Participation involves a one-time process in which you:

- Complete an electronic consent form on the CoTEDS Smart website
- Create a study account
- Connect your device (smartwatch, fitness tracker, or smartphone app) to the study platform
- If you own an Apple device, you will need to download an app to transfer the data.

£10 voucher as thank you for participation!

Sign up here!

If you have any questions, please contact: **coteds-project@kcl.ac.uk**



CoTEDS-Smart



What?

We are looking into understanding relationships between sleep, physical activity, mental health and physical health during parenthood and pregnancy by collecting passive data and combining it with other data collected in TEDS and CoTEDS.

We will then analyse the combined data to understand associations between fitness tracker data and other indicators of health and wellbeing



How?

Using smartwatches, fitness trackers and smartphone apps. Data will be collected and stored securely.

Why?

This may help to inform the development of better health support during pregnancy, the perinatal period and parenthood. Research questions we can address include:

- What signs and symptoms during pregnancy might distinguish between parents who experience perinatal physical/mental health difficulties
- How does parenthood change physical activity?
- How does the body and its systems change over the duration of pregnancy?



Data Transparency

What data will be collected as part of CoTEDS SMART?

We will collect past physical activity data from participants' wearable fitness trackers and combine it with health and wellbeing data collected in TEDS/CoTEDS. Past physical activity information that will be collected are as follows:



Apple Health

Active Energy

Exercise Minutes, steps and heights climbed

Heart Rate

Walking/Running and Cycling Distance

Workouts and durations

Respiratory Data

Sleep Analysis

OURA

Sleep, Readiness, and Activity Data

Heart Rate Data

Workout Data

Body Temperature

Sleep Stages

Activity Burn: Basal metabolic rate (BMR) vs. active calories.

GARMIN

Activities & Intensity

Steps and Distance

Daily Health Stats

Heart Rate Data

Stress Levels

Body Battery

Pulse Ox

fitbit

Heart Rate Data

Sleep: quality & interruptions.

Activity & Exercise: Steps, distance, calories.

How will my data be used, and what for?

Your fitness tracker data will be combined with questionnaire data from CoTEDS/TEDS to explore your:

- sleep
- activity
- wellbeing
- health during pregnancy and parenthood

Analyses will be carried out by approved academic researchers only.

Data will be analysed to understand associations between datasets and indicators of health and wellbeing. We are interested in whether fitness tracker data can provide us with additional insights that data collected by other means cannot, i.e., can fitness tracker data serve as an early warning of potential health problems).



Data Protection Act 2018

How will my data be protected?

Data is processed under UK data protection law, including the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018. The legal basis under which CoTEDS processes your data is the performance of a task in the public interest, namely scientific and statistical research purposes. Full details of our legal basis are provided in the TEDS Privacy Policy [here](#).

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Information Sheet for Participants

PLEASE SAVE A COPY OF THIS INFORMATION SHEET FOR YOUR RECORDS.

CoTEDS-Smart Pilot Study

Study summary

We would like to invite you to participate in the ***CoTEDS-Smart*** pilot study, which is an extension to the Children of Twins Early Development Study (CoTEDS). CoTEDS-Smart is focused on data that is passively collected via smartwatches, fitness trackers and smartphone health apps. Before you decide whether to take part, it is important for you to understand why the research is being conducted and what your participation will involve. Please take time to read the following information carefully and discuss it with others if you wish. Please feel free to contact the CoTEDS team using the details at the head of this document if there is anything that is not clear or if you would like more information. Thank you for taking the time to consider taking part in this study.

What is the purpose of this study?

At CoTEDS, we are interested in understanding relationships between (amongst other things) sleep, physical activity, mental health, and physical health during pregnancy and parenthood. To complement the questionnaire data that you so kindly provide us with, we are interested in exploring these relationships using data that is routinely collected by smartwatches, fitness trackers, and smartphone apps. Many people own and use such devices now, and sharing the data collected by them has the potential to provide exciting new scientific insights. For example, previous research has shown that fitness tracker data reflects the hormonal changes experienced during pregnancy.

Who can take part in CoTEDS-Smart?

Any TEDS twins and/or coparents enrolled in the Children of TEDS study with their child(ren) are eligible to take part. We're looking for people who use a smartphone, Fitbit, Apple Watch, Garmin or Oura Ring.

Do I have to take part?

Participation in CoTEDS-Smart is entirely optional; you can continue taking part in CoTEDS without joining CoTEDS-Smart. You can also join CoTEDS-Smart regardless of whether you have completed your CoTEDS questionnaires.

Once you have read the information sheet, please contact us if you have any questions that will help you decide about taking part.

If you decide to take part in this study, you will be asked to complete a consent form to confirm that you have read this information sheet and that you agree to us using your data in the ways stated.

What if I don't own a smartwatch or fitness tracker?

At present, you must already use a compatible device to take part. In the future we hope to be able to provide smartwatches or fitness trackers to some participants who don't already own one (this depends on funding). If you have expressed an interest in sharing smartwatch or fitness

tracker data, should we provide you with a device, then keep an eye on your emails as we may be in touch.

What will I be asked to do?

Participation involves a one-time process in which you:

- 1) Read this information sheet
- 2) Complete an electronic consent form on the CoTEDS Smart website
- 3) Create a study account
- 4) Connect your device (smartwatch, fitness tracker, or smartphone app) to the study platform
 - a) If you own an Apple device, you will need to download an app from <https://onsentia.com/> to transfer the data.

You do not need to wear your device more than you normally do.

How long will this take?

Completing all required steps should take no more than 35 minutes, including upload time. For many devices it will take much less than this.

What do I need in order to take part?

You will need:

- access to a smartphone, tablet, or computer
- login details for your device's health data platform (e.g., Apple Health, Fitbit, Garmin, Oura)

Step-by-step guidance

To share with us the data that your device collects you will need to connect your device on a secure and open research platform called 'RADAR-base' which is managed by Onsentia (developed at King's College London). From the website you can:

1. Check that you can take part.
2. Sign an electronic consent form.
3. Set up a study account.
4. Complete up to 4 tasks.

Connecting your device couldn't be simpler. Please see the device specific guide for more detailed information. We have designed a step-by-step process that will support anyone to connect their device in a stress-free way. Completing these steps should take no more than 35 minutes, except for Apple devices which can take longer to upload data via the Onsentia app.

What are the possible benefits of taking part?

If you participate, you will help us learn about relationships between physical activity, heart rate, sleep, mental health and physical health by sharing with us the high quality, accurate data collected by your fitness tracker. Ultimately this may help to inform the development of better health interventions, and improved treatment during pregnancy, the perinatal period and parenthood.

Are there any risks in taking part?

There are no foreseeable risks in participating in the study.

Rewards for participation

To thank you for taking the time to participate, we will email you a £10 Love2Shop voucher. These vouchers can be used at over 90 retailers including Sainsbury's, Asos, Zizzi, John Lewis, and Just Eat. For more information about where this voucher can be spent, please go to <https://business.love2shop.co.uk/rewards/digital-rewards/e-gift-cards/#spend>. Please make sure that you provide us with your email address on the consent form so that we can send you your vouchers. Please make sure that you provide us with your email address on the consent form so that we can send you your vouchers.

Data Protection and Confidentiality

How will my data be protected?

Your data will be processed under the terms of UK data protection law (including the UK General Data Protection Regulation (UK GDPR) and the Data Protection Act 2018). The legal basis under which CoTEDS processes your data is the performance of a task in the public interest, namely scientific and statistical research purposes. Full details of our legal basis are provided in the TEDS Privacy Policy at <http://www.teds.ac.uk/about-teds/privacy>.

What will you use this data for?

We will collect data from participants' wearable fitness trackers and combine it with other data collected in TEDS and CoTEDS. We will then analyse the combined data to understand associations between fitness tracker data and other indicators of health and wellbeing included in TEDS/CoTEDS. We are interested in whether fitness tracker data can provide us with additional insights that data collected by other means cannot.

There are many research questions that we can address using fitness tracker data. Some examples will include:

- Can fitness tracker data collected during pregnancy distinguish between parents who experience perinatal physical/mental health problems and those who do not? (i.e. can fitness tracker data serve as an early warning of potential health problems)
- Does fitness tracker data agree with self-reported questionnaire data on sleep quality and duration? Which measure is most predictive of health outcomes?
- Does parenthood lead to an increase/decrease in physical activity?

What identifiable information will be collected?

We may collect the following identifiable details to manage your participation:

- Name
- Email address
- Date of birth
- Postcode
- Electronic consent form

These details are held securely and separately from your research data.

What past physical activity information will be collected?

From Apple Health

- 1. Active Energy**
- 2. Exercise Minutes**

3. Flights Climbed

4. Heart Rate: Including resting heart rate, walking average, and heart rate variability (HRV).

5. Steps

6. Walking + Running Distance

7. Cycling Distance

8. Workouts: Specific activity labels (e.g., yoga, swimming, HIIT) and durations.

9. Oxygen Saturation (SpO2)

10. Respiratory Rate

11. Sleep Analysis: Detailed sleep stages (REM, Core, Deep) and total duration.

From Oura:

1. Sleep, Readiness, and Activity Data

2. Heart Rate Data: Continuous heart rate and HRV.

3. Workout Data

4. Body Temperature: Nocturnal skin temperature deviations.

5. Sleep Stages: High-resolution hypnogram data (Deep, Light, REM, Awake).

6. Activity Burn: Basal metabolic rate (BMR) vs. active calories.

From Fitbit:

1. Heart Rate: Real-time and resting heart rate, HRV.

2. Activity and Exercise: Steps, distance, and calories.

3. Sleep: Detailed metrics on sleep quality and interruptions.

From Garmin:

1. Activities

2. Daily Health Stats: General health summaries.

3. Heart Rate Data: Continuous and resting heart rate and HRV.

4. Stress Levels

5. Body Battery

6. Pulse Ox

7. Steps and Distance

8. Floors Climbed

9. Intensity Minutes: Breakdown of moderate vs. vigorous activity.

Things to know about sharing physical activity information

- If you have connected your iPhone/Fitbit/ Oura/Garmin to other fitness tracking devices and apps the same information (listed above) from those other devices and apps will be collected and used in our research.
- If you agree to share Fitbit, Garmin or Oura ring information: Once you have provided permission to share your information, you do not need to do anything further.

For your information: It takes approximately 9 days for the research team to receive your physical activity information from Fitbit/Oura/Garmin. In some circumstances, e.g. if there is a connection issue, it may take up to 4 weeks for the team to receive your data. This is because Fitbit/Oura/Garmin will send the information to us in small chunks rather than all in one go. If there are gaps in the physical activity data we receive from

Fitbit/Oura/Garmin, we will make one further request to Fitbit/Oura/Garmin to download your data.

- You will share data which is stored on your fitness tracker or health app on your smart phone. The amount of data you share will depend upon your device but may be several years of data. We work within strict ethical guidelines meaning we will only collect the information you have agreed to share with us.
- If you agree to share information from an Apple watch and have family sharing: under certain circumstances e.g. use of subscription services, your forename maybe associated with the device name. If we receive your forename with the name of your device, we will delete it as soon as possible.

How will my information be used?

Your fitness tracker data will be combined with questionnaire data from CoTEDS/TEDS to explore your:

- sleep
- activity
- wellbeing
- health during pregnancy and parenthood

Analyses will be carried out by approved academic researchers only.

Will my data be shared?

Your data collected via CoTEDS-Smart, along with other data you have previously provided, will **not** be shared with anyone other than the data collection company Onsentia and academic researchers who follow the typical TEDS data access protocol. This data will not be used for any purpose other than health/wellbeing research. Your data will only be shared in a form in which all personally identifiable information is removed, so that you and other individual participants cannot be identified.

How long will you keep my data?

As CoTEDS is an ongoing longitudinal study, the research data will be kept for the duration of the study. When data collection for CoTEDS eventually ends (we don't know when this will be, probably many years from now), the research data will be archived for future research purposes. Data from questionnaires will be stored in an anonymous form whereby participants are not identifiable (e.g. administrative data such as names and contact details will be deleted).

Data Protection Statement

If you would like more information about how your data will be processed under the terms of UK data protection laws, please visit the link below:

<https://www.kcl.ac.uk/research/support/research-ethics/kings-college-london-statement-on-use-of-personal-data-in-research>

What if I change my mind about taking part?

You have the right to ask us to withdraw your data. If you have any objections to us using your data for further research in the future, just contact us using the details at the head of this document and we can arrange the removal of your data.

Will my taking part be kept confidential?

All information is confidential, and the data is used anonymously. We are not allowed to pass any information to outsiders such as police, doctors, or insurance companies. At the end of the study, the data might be used in future research that is ethically approved.

How is the project being funded?

CoTEDS and CoTEDS-Smart are funded by the Wellcome Trust. CoTEDS-Smart is additionally funded by DATAMIND. CoTEDS and CoTEDS-Smart are independent of any commercial interest and is concerned with research only.

Any questions?

If you have any questions, please do not hesitate to contact CoTEDS by email using the details at the head of this document.

Ethical Clearance for this study has been received from King's College London Research Ethics Committee (Reference Number: 1935)

If you feel this study has harmed you in any way or if you wish to make a complaint about the conduct of the study you can contact King's College London using the details below for further advice and information:

Research Ethics Committee: The Chair, Health Faculties Research Ethics Subcommittee, rec@kcl.ac.uk

You can also contact the KCL Data Protection Officer, whose contact details are given at the bottom of the KCL privacy notice (<https://www.kcl.ac.uk/terms/privacy>). If you are not satisfied with the response given by the KCL Data Protection Officer, then you are entitled to complain to the Information Commissioner's Office (<https://ico.org.uk/global/contact-us/>).

Thank you for reading this information sheet and for considering taking part in this research.